

Tunisia, audited — the complete audit

Five absences. Nine legs. One metabolism. Every number recomputed in code, every correction applied into the design, every claim graded.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The five absences and the one metabolism

Five structural absences — monetary sovereignty (debt \$40.5B, ~79% of GDP, 26.2% of export revenues to service), sovereign compute (InstaDeep: founded in Tunis, sold for \$680M), strategic materials (phosphogypsum at Gabès costing the sea >\$58M/yr), water, nutrition — answered by nine cash-positive legs on one metabolism of power, water, data and workforce. The audit is finished; the corrections are in the design.

2. The FX bridge — every dollar, by leg

Each leg below is volume × price × utilization, with the double-counting checks applied — the design carries the checked ledger.

LEG	TYPE	VOLUME	PRICE	\$M/YR	GRADE
Strategic metals (U + heavy REE, acid stream)	gross export	716,500 lb U3O8 + 585 t REO	\$80-95/lb U3O8 · \$30/kg REO	\$75-86M	STRONG
Solar fuel substitution (500 MW)	import substitution	745 GWh/yr displaced (876 GWh less 131 GWh to H2)	\$142/MWh (gas \$12/MMBtu x 11 MMBtu/MWh + \$10/MWh O&M); government anchor: 1,100 GWh ≈ \$125M/yr gas saving for the same 500 MW fleet (Pg14)	\$95-116M	PLAUSIBLE
Hydrogen → ammonia (loop-internal offtake)	import substitution	2.63 kt H2/yr → 14.9 kt NH3 for the GCT chain	\$500/t imported ammonia displaced (external offtake upside: \$8-9.5/kg H2)	\$7-25M	SPECULATIVE
Terfas (desert truffles)	gross export	940 t/yr mid (2,500 ha x 376 kg/ha; 70% exported)	EUR20-40/kg farm-gate	\$14-29M	PLAUSIBLE
Food import substitution	import substitution	1% of the \$3.0B food import bill	imported protein \$15-25/kg vs local \$5-10/kg	\$30-30M	SPECULATIVE

LEG	TYPE	VOLUME	PRICE	\$M/YR	GRADE
		(frame; volume bridge is a pilot deliverable)			
Compute colocation services	gross export (services)	colo park, tenant-dependent	market colocation rates	\$15-15M	<i>SPECULATIVE</i>
ESG financing delta	financing delta (not FX)	100-150bp on ~\$800M portfolio debt	interest saved, not foreign exchange	\$8-12M	PLAUSIBLE
Total external-account effect (mid)				~\$279M/yr	\$80M strong · \$137M plausible · \$61M speculative · \$0 verified

By kind: exports \$117M · import substitution \$152M · financing delta \$10M (interest saved, honestly labeled as not-FX). **By grade:** \$80M strong · \$137M plausible · \$61M speculative · \$0 verified — nothing is commercially validated yet, and that ledger reads zero on purpose.

3. The energy ledger — every kWh, by leg

LEG	GWH/YR	THE FORMULA
Pump-as-turbines (return flows)	3.9	$70\% \times 150\text{k m}^3/\text{d} \times 50\text{ m} \times 75\% \times 8,760\text{ h}$
Heat (kiln + exotherm)	30	9% of 333 GWh envelope
Flexibility (load shift)	20	dispatch value — not recovered joules
CO ₂ ; mineralized	30 kt	mass, not energy
Total	33.9 recovered + 20 dispatch	isobaric ERD inside the 4 kWh/m³ benchmark — never double-counted

4. Phase 0 — every dollar, itemized

ITEM	COST	WHAT IT BUYS
Water panel: 100 points, full suite (F, Cd, Pb, Cr, As, salinity, nitrate)	\$35,000	the first anchor measurement
Acid-stream assays: 12 points (U, REE, Ra-226, Po-210, full metals) + GCT access	\$40,000	the second anchor measurement
Emissions inventory: 30 passive samplers (PM _{2.5} /SO ₂ /fluorides) + lab	\$25,000	baseline for the health scoreboard
Cohort protocol: 3,000-family registry, ethics, data platform v0	\$35,000	the DALY ledger becomes MEASURED

ITEM	COST	WHAT IT BUYS
Health pilot first tranche: 100 households + 2 schools, 90 days	\$58,000	filters, clean-air boxes, KPIs
Legal + finance: health fund, industrial SPV, land-lease pre-agreements, tender registration	\$25,000	structure before scale
Engineering + project management	\$35,000	the 90 days are run
List price	\$253,000	list price \$253k; the two anchor measurements cost \$75k and land first — the claimed range is the phased-execution plan, not the list

5. Every claim, graded

CLAIM	GRADE	WHY
\$40.5B debt (~79% GDP); 26.2% of export revenues to service	STRONG	MEASURED via World Bank API on 2026-09-09: debt \$40.46B, debt/GDP 78.7%, service 26.17% of export revenues (2024)
\$3.0B food and \$4.9B fuel imports	STRONG	fuel \$4.895B verified on WITS/Comtrade (HS27, 2024); food per the WB food classification
InstaDeep founded in Tunis, sold for \$680M	STRONG	public deal record; the emblem for talent without infrastructure
Gabès: >\$58M/yr of REE value lost to the sea	STRONG	Science of the Total Environment (2021): up to 1,523.67 t/yr REE, ~\$58M/yr losses
U + heavy REE from the acid stream, ~\$80M/yr	STRONG	flowsheet industrial since 1978 (Freeport), patent expired, Morocco building it now; volume derived from cited primitives; Gafsa-specific assays are Phase 0
~\$285M/yr external-account effect, tiered	TIERED	\$80M strong (proven flowsheet/cited anchor) · \$137M plausible (gated) · \$68M speculative (assumed volume/price)
33.9 GWh/yr recovered + 20 GWh/yr dispatch value	DERIVED	every leg computed from cited primitives; dispatch value honestly labeled — not joules
\$2,592 per healthy year (water rung)	PLAUSIBLE	arithmetic DERIVED in code; DALY anchors transferred from Iranian studies — the cohort baseline makes it MEASURED
Phase 0 = \$150-250k, itemized	PLAUSIBLE	itemized list price \$253k; the two anchor measurements cost \$75k and land first
Commercially validated	ZERO	no buyer has signed anything — published as zero on purpose

6. The hidden dependencies

DEPENDENCY	TYPE	STATUS	NOTE
Desal brine permit	regulatory	in design	Gabes outfall / diffuser; permit required before SWRO FID

DEPENDENCY	TYPE	STATUS	NOTE
PG chemistry (Ra-226 partitioning)	technical	Phase 0 assay	flowsheet keeps radium in gypsum (Freeport precedent); re-verify on Gafsa stream
Uranium handling	regulatory	gate	CNSTN licensing, NORM transport, export controls
REE separation capex	capital	later phase	+\$244M solvent-extraction plant beyond the ~\$100M circuit
Solar bankability	financial	gate	DSCR ~0.9-1.0 at the record-low tariff; bankable region or grant support required
Grid connection	infrastructure	later phase	3 GW connection + 400 kV upgrades ~\$450M
Compute accelerators	regulatory	gate	export-control licenses; colocation until an anchor tenant is licensed
Truffle yield	agronomic	gate	376 kg/ha is the 20-yr Murcia average; gate = 3-season record ≥ 200 kg/ha
Export prices	market	exposure	U spot \$80-95/lb; truffle EUR20-40/kg — no contracted offtake yet
Financing	financial	uncommitted	DFI channel modeled for solar/desal; nothing committed

7. The structure and the pilot/capital separation

The design is six independently bankable modules sharing infrastructure. The loop is the outcome, never the precondition — no module's economics depends on another module succeeding, and every gate is public. World Bank CCDDR (2023): Tunisia's water-sector investment need ~\$17.1B and energy-sector ~\$34.7B through 2050. The page is not that program and does not claim to be: Phase 0 is \$150-250k of measurements, and the whole core build is ~\$250-650M phased over years 2-5, module by module, each gated. The CCDDR numbers confirm the scale of the problem; they do not compete with a pilot gate.

Africa Energy Portal (2026): 500 MW of utility-scale solar approved (~\$400M investment). Same class as the page's P3 module (500 MW) — the record-low ~\$31/MWh awards are the reference the DSCR gate is tested against.

8. The Fed transmission — measured, not restated

World Bank API, measured 2026-09-09 (Pg1): external debt **\$40.46B, 78.7% of GDP**; debt service **26.2% of export revenues**; reserves **\$9.34B** = 4.3 months of imports; food \$3.00B + fuel \$4.90B of imports; CA -1.5% of GDP; growth 2.5%, inflation 5.2% (2025).

Channel 1 — the debt trap: the “original sin” (Pg12) — no long-term borrowing in the dinar abroad — exposes the state to the Fed cycle (Pg13): Fed hikes lift global yields, refinancing turns expensive, and the 26.2% of export revenues that debt service absorbs compounds while reserves (4.3 months) drain.

Channel 2 — imported inflation: the rate differential pulls capital to US safe assets, the dinar depreciates, and the dollar invoice on food (\$3.00B) and fuel (\$4.90B) inflates. The BCT — autonomous, non-monetizing under Organic Law 2016-35 (Pg4/Pg5) — must hold rates to defend the dinar: correct orthodoxy, strangled growth.

The loop's answer: owned export engines earn ~\$117M/yr of gross exports outside the Fed cycle; the substitution legs take \$152M/yr out of the dollar invoice directly; reserves rise, the rating runs B-/Caa1 → BB-, and the refinancing premium shrinks for the state with its own hard-currency engines. Not defiance of the Fed — independence from the channel.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

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66. [H12] Drieling 2022 RCT (75 asthmatic children): HEPA in bedroom+living room -> poorly-controlled asthma IRR 0.43-0.45; unplanned clinical utilization IRR 0.35 — <https://pubmed.ncbi.nlm.nih.gov/34980119/...>
67. [H13] Spirulina RCT (Cambodia, n=179): 2 g/day x 50 days -> anemia reduction significant (p=0.004); weight-gain trend p=0.07 (do not overclaim growth) — <https://pubmed.ncbi.nlm.nih.gov/36476193/...>
68. [H14] EJAtlas: Gafsa affected population 30,000-70,000; cancer/asthma/infertility reported high; conflict since 2008 — <https://ejatlas.org/conflict/phosphate-mining-in-gafsa...>
69. [H15] NRGi: 'serious dearth of rigorous studies' linking phosphate operations to health outcomes; 7% birth-defect claim is activist-reported, unverified — <https://resourcegovernance.org/articles/uncovering-impacts-phosphate-mining-tunisian-women...>
70. [H16] Geographical 2024: asthma onset age 2 documented (Redeyef); nearest hospital 70 km; doctors prescribe leaving Gafsa — <https://geographical.co.uk/science-environment/gafsa-silent-suffering-the-hidden-cost-of-phosphate-mining-in-...>